

Activity

1. CCTV image suffers salt-and-pepper noise due to transmission errors.

Technique Used: Median filter

Reason: Median filtering removes black white noise pixels affecting white preserving the edges. It improves the image quality without blurring important data.

2. A satellite image is corrupted by Gaussian noise from sensors.

Technique Used: Gaussian filter.

Reason: Gaussian filtering smooths image and reduces Gaussian noise caused by sensors. It produces clearer image for further analysis.

3. A medical X-ray image has random noise but fine edges must be preserved.

Technique Used: Bilateral filter.

Reason: Bilateral filtering removes noise while preserving important features helps doctors view bones tissues more clearly.

4. A Scanned document has black and white pixels scattered randomly.

Technique Used: Median filter.
Reason: Median filter removes salt and pepper noise without affecting the text. It improves the readability of the scanned document.

5. A low-light image shows heavy grain without clear boundaries.

Technique Used: Non-local Means (NLM) denoising.

Reason: NLM removes heavy grain noise while preserving image details and edges. It produces a cleaner image with better visual quality.

6. A foggy road image has very low contrast.

Technique Used: Histogram Equalization.
Reason: Histogram equalization is an image contrast enhancement technique that makes the image intensity values more visible in low contrast conditions.

7. An underwater image appears dark and washed out.

Technique Used: Contrast stretching

Reason: Contrast is stretching enhances brightness and color contrast. It makes underwater objects appear clear and more visible.

8. A chest X-ray shows poor visibility of internal structures.

Technique Used: CLAHE (Contrast Limited adaptive Histogram equalization)

Reason: CLAHE improves local contrast without over-enhancing noise.

9. A night-time traffic image lacks intensity variation.

Technique Used: Histogram equalization

Reason: Histogram equalization increases intensity variation and enhances visibility of vehicles and roads at night.

10) A grayscale image uses only a narrow intensity range.

Technique used: Contrast stretching (Normalization)

Reason: Contrast stretching expands the narrow intensity range to the full range, making the image clearer and improving overall contrast.